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Financial inclusion and economic development in selected east African countries: A human development index approach

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ABSTRACT

While Gross Domestic Product (GDP) per capita remains a common indicator of economic progress, it often fails to accurately reflect a population's true well-being. Countries with similar income levels can exhibit significant disparities in human development, as measured by the Human Development Index (HDI). The existing literature emphasizes the influence of financial inclusion on economic growth, but overlooks its broader implications for development. It also focuses on conventional metrics, such as bank branches, Automated Teller Machines (ATMs), and outstanding loans. Relying on traditional measures while neglecting recent Information and Communication Technology (ICT) advancements and monetary policy instruments, such as interest rate spread, poses a challenge for effective policy implementation. This study extends financial inclusion indicators to empirically assess their impact on shaping development outcomes in developing countries, such as those in East Africa. A suite of econometric techniques was employed to analyze the data, including panel unit root, cointegration, and cross-sectional dependence tests. Additionally, the Fully Modified Least Squares (FMOLS) and Dynamic Ordinary Least Squares (DOLS) methods have been employed to estimate the model coefficients. The findings reveal how financial inclusion significantly enhances the ability of developing countries to translate economic growth into human development. The study concludes by recommending policy interventions to expand financial access and strengthen institutional frameworks to support inclusive development pathways.

1. Introduction

Financial inclusion, defined as the availability and accessibility of affordable financial services to all individuals and businesses, is increasingly recognized as a key driver of economic growth and poverty reduction globally(Ediagbonya & Tioluwani, 2023; Ozili, 2021; H. T. T. Tran & Le, 2021). According to Demirgüç-Kunt et al. (2022), over 1.4 billion adults were financially excluded in 2021, with the majority residing in developing countries. The importance of financial inclusion is highlighted in the United Nations' Sustainable Development Goals (SDGs), precisely Goal 8, which aims to promote inclusive and sustainable economic growth, employment, and decent work. It is widely acknowledged that access to financial services can empower individuals, especially the poor, to save, invest, and manage risk, which in turn fosters economic stability and growth (Demirgüç-Kunt et al., 2022; Pazarbasioglu et al., 2020).

Increased financial inclusion is also linked to broader financial sector development. Financial institutions, including banks, credit unions, and microfinance institutions, play a crucial role in mobilizing savings, allocating credit, and facilitating payments, all of which are vital to economic growth and development. Studies have demonstrated that greater access to financial services can spur entrepreneurial activity (Frimanslund et al., 2023), reduce income inequality(Demir et al., 2022), and promote long-term investment(Chen et al., 2021; Hasan et al., 2021; Pradhan et al., 2021). However, despite the global push for financial inclusion, significant disparities remain across regions, particularly in Sub-Saharan Africa, where a large portion of the population remains financially excluded.

Most of the literature focuses on the impact of financial inclusion on economic growth and ignores economic development in general (SARMA, 2011). In addition, the majority of the literature concentrates on classic financial inclusion metrics such as the number of bank branches and ATMs available per 100,000 population and outstanding loans(Hussain et al., 2024). The reliance on traditional financial

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inclusion metrics while overlooking recent advancements in financial services driven by information and communication technology (ICT) poses a significant challenge to the effective implementation of policy. The relationship between financial inclusion and economic development in these regions requires scientific exploration, mainly through specific financial inclusion indicators such as mobile cellular subscriptions (MCS), interest rate spread (IRS), domestic savings (GDS), and credit (DCPS), as well as economic development indicator that is the Human Development Index (HDI).

East Africa has made significant progress in advancing financial inclusion; however, the levels of access to financial services and their impact on economic development vary considerably across the region. Financial inclusion in East Africa is characterized by a mix of formal and informal financial systems, with mobile banking playing a particularly transformative role in countries like Kenya, Rwanda, Uganda, and Tanzania. According to the Global Findex Database (Demirgüç-Kunt et al., 2022), more than 65 % of adults in East Africa have access to some form of financial service, primarily driven by mobile banking platforms like Kenya's M-Pesa and mobile money in Rwanda. However, traditional financial institutions such as commercial banks still play a critical role in driving economic growth by providing credit and savings services to the private sector (Gizaw et al., 2024a).

The role of financial inclusion in East Africa's economic development is crucial and cannot be overstated. Countries in the region have experienced substantial economic growth over the past two decades, primarily driven by the agricultural and services sectors. Access to financial services is crucial in these sectors, enabling farmers and small businesses to invest in productivity-enhancing technologies and infrastructure (Njoroge, 2020; Otieno, 2017). Studies have shown that improving financial inclusion can significantly increase investments in small and medium-sized enterprises (SMEs), which are the backbone of most East African economies(Otieno, 2017). Moreover, financial inclusion provides households with tools to manage economic shocks, reduce income inequality, and build financial resilience, all of which contribute to long-term economic growth(Pomeroy et al., 2020; Ratnawati, 2020).

Despite progress in financial inclusion, East Africa faces significant challenges, particularly in rural areas where formal banking services are limited and financial literacy remains low. Rwanda has increased its financial inclusion rate to 86 %, driven by the expansion of digital finance and microfinance. However, domestic credit to the private sector (DCPS) remains low at 29.9 % of GDP, with gross savings (GS) at 13.9 % (WB, 2024). Uganda's financial inclusion rate reached 66 %, but its DCPS is one of the lowest in East Africa, at 12.5 %, with only 8.4 commercial bank branches (CBBs) per 100,000 adults and a DGS rate of 17.1 %. Tanzania's financial inclusion rate is 59 %, but DCPS remains low at 11.8 %, with 6.3 CBB per 100,000 adults and a GS rate of 14.6 %. Kenya leads the region with an 82 % financial inclusion rate, 31.4 % DCPS, 14.8 CBB, and the highest GS at 19.2 % (WB, 2024). However, disparities in credit access and banking infrastructure persist, especially in rural areas (Benami & Carter, 2021; Friedline et al., 2020; T. Yang & Zhang, 2022).

Economic growth in East Africa is crucial for promoting regional development, alleviating poverty, and achieving sustainable economic development. Financial inclusion, ensuring widespread access to essential financial services, has proven to be a significant contributor to economic growth by promoting capital accessibility(Gizaw et al., 2024a; He & Yoo, 2024), improving resource allocation(Chatterjee, 2020; Pazarbasioglu et al., 2020), and fostering business development(Chen et al., 2021; Ekechi et al., 2024). Empirical research consistently demonstrates a positive relationship between financial inclusion and economic growth. For example,(Demirgüç-Kunt et al., 2017; Q. Tran et al., 2021) reported that a 1 % increase in domestic credit to the private sector correlates with a 0.2 % increase in GDP growth in developing regions, illustrating the significance of credit access for business expansion and economic activity.

Existing research has explored the link between financial inclusion with indicators such as savings, credit, bank branches, ATMs, the number of people having a bank account, and economic growth represented by Gross Domestic Product (GDP) (Amoah et al., 2024; Fuseini et al., 2024; He & Yoo, 2024). Nevertheless, a comprehensive understanding of the relationship between economic development proxied by the Human Development Index (HDI) and specific financial inclusion indicators such as mobile cellular subscriptions (MCS), interest rate spread (IRS), gross domestic savings (GDS), and domestic credit to the private sector (DCPS) remains lacking within the context of East African countries. The choice of HDI as an independent variable over GDP per capita growth has been motivated by various reasons.

Unlike GDP, which focuses solely on economic output and income, HDI provides a multidimensional perspective by incorporating health (life expectancy), education (mean and expected years of schooling), and income (GNI per capita). This aligns with the broader scope of economic development targeted by the study. Additionally, HDI captures the societal and structural changes driven by financial inclusion, including improved access to education and healthcare. HDI also reflects the multidimensional nature of development as outlined in the SDGs, making it a more suitable metric for analyzing long-term and sustainable impacts. Furthermore, existing literature demonstrates that financial inclusion significantly influences components of HDI, including education and health outcomes(SARMA, 2011; Sharma, 2016). This validates HDI's relevance to the study's objectives. The findings of this research will provide policymakers with valuable insights, enabling them to formulate effective strategies that enhance financial access and promote sustainable economic development across East Africa.

2. Literature review

2.1. Theoretical literature review

The finance-growth nexus, endogenous growth, savings-led growth, and the McKinnon-Shaw Hypothesis are key theoretical frameworks for this study. The finance-growth nexus theory posits a causal relationship between financial development and economic growth, suggesting that an advanced financial system can enhance capital allocation, risk management, and innovation, thereby stimulating economic activity. Endogenous growth theory emphasizes technological progress and human capital as key drivers of sustained economic growth, with financial development playing a crucial role in supporting these elements. The savings-led growth theory emphasizes the importance of savings for investment and growth, while the McKinnon-Shaw Hypothesis suggests that financial liberalization promotes higher savings rates, increased investment, and ultimately, accelerated growth.

The finance-growth nexus, a cornerstone of economic theory, posits a causal relationship between financial development and economic growth(Schumpeter, 1912). A substantial body of research has empirically supported this proposition, encompassing historical analysis, cross-country regressions, and within-country studies. Economic historians have meticulously examined the role of financial revolutions in driving economic growth. For instance, studies on seventeenth-century Holland(Neal, 1990; Vries & Woude, 1997), eighteenth-century England(North & Weingast, 1989), and the nineteenth-century United States(Rousseau & Sylla, 2004; Sylla, 2006, 2008) have demonstrated how financial innovations facilitated rapid economic expansion and global dominance. Conversely, the absence of such financial revolutions has been linked to slower economic progress in emerging economies, such as nineteenth-century Russia(Anan'ich, 1999), Argentina(Davis & Gallman, 2001), and early twentieth-century Mexico(Haber, 1991, 1997; Haber et al., 2003; Maurer, 2002). Comparative historical studies further underscore the strong correlation between financial innovation and economic development (Afjal, 2023; Gruin & Knaack, 2020; Hossain et al., 2024; Hussain et al., 2024).

Complementing historical analysis, cross-country regression studies have further solidified the finance-growth nexus. Goldsmith (1969) and McKinnon (1973) pioneering work established a link between financial

system development and economic performance. King & Levine (1993) formalized this relationship, demonstrating a strong correlation between banking sector development, physical capital accumulation, and economic growth across over 80 countries. Their findings suggest a causal relationship where financial development precedes and drives economic growth. Levine & Zervos (1998) extended this research to incorporate stock market development, revealing that both stock market liquidity and banking sector development are positively associated with current and future economic growth.

To address concerns regarding unobserved heterogeneity and spurious causality, Rajan & Zingales (1998) provided compelling evidence that industries reliant on external finance exhibit faster growth in countries with more developed financial systems. Wurgler (2000) further strengthened this argument by demonstrating that financially developed countries allocate investments more efficiently. Fisman & Love (2004) This research was extended by investigating the impact of financial development on industry-level growth. It found that finance-dependent industries tend to grow more rapidly in countries with robust financial institutions over the long term.

Studies exploiting within-country variation, such as those by Jayaratne & Strahan (1996) and Black & Strahan (2002), further corroborated these findings, demonstrating that regulatory reforms to increase bank competition stimulated economic growth and entrepreneurial activity. Recent research has delved deeper into the specific mechanisms underlying the finance-growth nexus. Guisoet al. (2004) found that financial development in Italy fostered entrepreneurship and firm growth, thereby contributing to accelerated GDP per capita growth. Similarly, Appiah et al. (2023), J. Yang and Ni (2022), and other researchers have shown that financial development can reduce industrial concentration and boost productivity growth.

Alternatively, endogenous growth theory posits that a well-developed financial sector can significantly contribute to economic growth. A robust financial system can foster sustained economic expansion by enhancing the allocation of savings, stimulating innovation, and improving investment efficiency. Hassan et al. (2011) argue that as economies grow, the demand for financial services increases, leading to the expansion of the financial sector itself. Pradhan & Sahoo (2021) identify four primary hypotheses to explore the relationship between financial sector development and economic growth: the supply-side hypothesis, the demand-side hypothesis, the feedback hypothesis, and the neutrality hypothesis.

The supply-side hypothesis posits a unidirectional causal relationship where financial development drives economic growth (Gregorio & Guidotti, 1995). Studies by Cheng et al. (2021) and Ibrahim & Alagidede (2018) support this view, arguing that a well-developed financial sector can improve resource allocation, mitigate risks, enhance corporate governance, and provide efficient pricing signals, fostering inclusive economic growth. Luintel & Khan (1999) provide empirical evidence for this hypothesis using VAR estimates from 10 developing countries between 1950 and 1990. Further support comes from studies in various regions, including Taiwan, South Asia, and OECD countries (Akinci et al., 2014; Bojanic, 2012; Chang & Caudill, 2005; Ray, 2013; Wadud, 2009). In contrast, the demand-side hypothesis suggests that economic growth precedes financial development (Patrick, 1996). Polat et al. (2015) argue that the growth of the enterprise sector stimulates the development of the financial sector. Kuznets (1955) noted a correlation between financial development and the economic cycle, with financial development accelerating as economies mature. Arestis & Demetriades (1997) distinguished between bank-based and market-based financial systems, suggesting that market-based systems are more closely linked to short-term growth. Zang & Kim (2007) found a positive correlation between financial development and economic growth in a sample of 74 countries from 1961 to 1995. Studies by Baliamoune-Lutz (2003) in Algeria and Egypt, Liang & Teng (2006) in China, and Khan (2010) in Pakistan provide further empirical support for this hypothesis.

Besides, the feedback hypothesis proposes a bidirectional causal relationship between financial development and economic growth. Abu-Bader & Abu-Qarn (2008) and Shan (2005) provide empirical evidence for this view. While Luintel & Khan (1999) found a unidirectional relationship between financial development and economic growth, Khalifa Al-Yousif (2002) found mixed results depending on country-specific economic policies. Wolde-Rufael (2009) confirmed the feedback hypothesis in Kenya, and Akinci et al. (2014) extended the debate to OECD countries. However, Cheng et al. (2021) rejected this hypothesis in their study of 72 countries. Finally, the neutrality hypothesis suggests that there is no causal relationship between financial development and economic growth. Lucas (1988) argued that financial development does not necessarily lead to economic growth. Solow (1956) and Swan (1956) proposed that economic growth eventually stagnates, regardless of further increases in financial capital. Atindéhou et al. (2005) found a weak causal link between the two variables. Studies by Kar et al. (2011) for 15 MENA countries and Ali (2014) for Pakistan also supports this view.

Another theory that explains the relationship between financial inclusion and economic growth is the "Savings-Led Growth Theory." This theory suggests a strong positive relationship between savings rates and long-term economic growth(Mohan, 2006). Higher savings rates can fuel business expansion, infrastructure development, and productivity growth by providing the necessary capital for investment. Financial inclusion, particularly access to savings instruments, is crucial in mobilizing these savings for productive use(Demirgüç-Kunt et al., 2008). Increased savings can reduce reliance on external borrowing, especially in developing economies with limited domestic capital (Odhiambo, 2009). Moreover, interest rates can influence savings behavior, with higher rates incentivizing savings and potentially boosting investment(Modigliani, 1986). Empirical evidence from rapidly growing economies like China and South Korea supports this theory, demonstrating a positive correlation between high savings rates and economic growth(Wang & Yao, 2003).

However, critics argue that savings alone are insufficient for sustained economic growth. Efficient financial systems, sound governance, and a skilled workforce are also essential for channeling savings into productive investments (Arestis & Demetriades, 1997). In conclusion, while savings are a critical component of economic growth, their impact is contingent upon effective investment allocation and supportive economic policies, particularly in developing economies like those in East Africa(Ncube, 2015).

The McKinnon-Shaw Hypothesis, introduced by Ronald McKinnon and Edward Shaw in the 1970s, postulates that financial repression, characterized by government-imposed interest rate ceilings and restrictive banking practices, can impede economic growth by limiting the availability of financial resources for investment(McKinnon, 1973; Shaw, 1973). The hypothesis posits that financial liberalization, which enables interest rates to reflect the actual cost of capital, can lead to increased savings, financial deepening, and ultimately, higher investment and economic growth. By enhancing the mobilization of savings and the allocation of resources to productive sectors, financial liberalization can stimulate economic growth, particularly in developing economies where financial inclusion is crucial (Khan, 2010). Empirical evidence supports this hypothesis, demonstrating a positive correlation between financial liberalization, higher savings rates, and economic growth in various countries (Arestis & Demetriades, 1997; Ndlovu, 2015).

2.2. Empirical literature review

Empirical studies examining the relationship between financial inclusion indicators and GDP growth in various countries have yielded mixed results. While many studies have found a positive correlation, others have reported either a non-significant or a negative relationship. Eton et al. (2019) investigated the impact of financial inclusion on economic growth in specific districts of Western Uganda. Their findings suggest that financial inclusion can contribute to equitable growth,

improve livelihoods, increase capital formation, and empower individuals to make informed financial decisions. However, they highlighted challenges such as limited financial literacy, inadequate infrastructure, and high transaction costs that may hinder the full realization of the benefits of financial inclusion. Fuseini et al. (2024) re-examined the moderating role of ICT in the nexus between financial development and banking efficiency in Africa. They find that the increasing number of Individuals using the Internet enhances the banking sector's efficiency. In contrast, fixed telephone subscriptions, mobile cellular subscriptions, and secured internet servers impact banking sector efficiency insignificantly. He and Yoo (2024) affirm that financial development has a positive influence on investment performance. They point out that changes in financial development do not substantially impact investment in the short term, and the marginal effect of financial development on investment is more pronounced in low— and middle-income countries.

Twesige & Uwamahoro (2015) their study investigated the relationship between financial inclusion and socioeconomic development in Musanze District, Rwanda. It examined how access to financial services, such as savings, loans, and mobile banking, influences economic outcomes, including income levels, employment, and poverty reduction. The findings revealed a strong positive correlation between financial inclusion and socioeconomic development. However, the study also highlighted challenges such as limited financial literacy and accessibility, which may hinder the full realization of the benefits of financial inclusion. Wani et al. (2024) explored the complementary interaction between financial inclusion and gender equality on economic growth in 48 countries. Their findings show that innovations in financial inclusion, combined with greater gender equality, have a multiplier effect on economic development.

Nkwede (2015) examined how access to financial services, such as banking, credit, and insurance, influences economic indicators, including GDP growth, poverty reduction, and employment rates. The findings revealed a significant positive relationship between financial inclusion and economic growth, indicating that increased access to financial services contributes to enhanced economic performance and poverty alleviation. However, the study also highlighted the challenges of financial exclusion, particularly in rural areas, and emphasized the need for policies to promote inclusive financial systems across Africa. In contrast, Gizaw et al. (2024b) assessed Sectoral allocations of domestic credit and their effects on economic growth in Ethiopia. They find that in the short term, credits to the public and industrial sectors negatively affect output growth, while credits to the agricultural, service, and private sectors have a small positive impact. Overall, credit allocation harms growth in the long term, suggesting that credit misallocation has more significant adverse effects over time than in the short term.

Williams et al. (2017) investigated the impact of financial inclusion on economic growth and poverty alleviation in developing countries, focusing on Nigeria. The study findings revealed a negative correlation between commercial bank depositors and real GDP growth, a significant gap in active ATM users between rural and urban areas, a positive correlation between commercial bank branches and real GDP growth, and a significant positive impact of government expenditures on real GDP growth. Odhiambo (2005) investigated the causal relationship between financial development and economic growth in Tanzania. The study findings revealed a bidirectional causal relationship, suggesting that both variables mutually reinforce each other in Tanzania. Kapaya (2020) also investigated the relationship between financial development and economic growth in Tanzania using time-series data. The study employed the Autoregressive Distributed Lag (ARDL) model and the bound testing approach to examine both the short-run and long-run dynamics. The study found a positive short-term relationship between financial system depth and economic growth, but a significant negative long-term relationship between financial system liquidity, efficiency, and economic growth. These findings suggest that while financial reforms and liberalization have had some positive short-term effects, they have not yet yielded the intended long-term positive impact on economic growth.

Mikebanyi & Kigabo (2021) investigated the relationship between financial development and economic growth in Rwanda over a 38-year period. Employing the Autoregressive Distributed Lag (ARDL) model and Granger causality test, the study examined both the short-term and long-term effects of financial development indicators, such as credit to the private sector and broad money supply, on Rwanda's economic growth. The findings revealed a significant positive long-run relationship between financial development and economic growth. Additionally, the Granger causality test indicated a bidirectional causal relationship, suggesting that financial development and economic growth mutually influence each other. Musembia & Chun (2020) analyzed the relationship between financial development, financial inclusion, and economic growth in Kenya using data from 1980 to 2018. Their findings indicate a significant positive long-run relationship, showing that financial development and inclusion are key drivers of economic growth. The study emphasizes that expanding access to banking and credit services further stimulates growth, underscoring the need for policies that promote financial sector development and inclusion to sustain Kenya's economic progress.

While the empirical studies mentioned have made notable contributions, several gaps remain. Most of these studies assessed financial development, overlooking some important indicators, such as the interest rate spread (the lending rate minus the deposit rate) and mobile cellular telephone subscriptions (MCS). Interest rate spread (IRS) plays a pivotal role in financial inclusion(Allen et al., 2016; Chirwa & Mlachila, 2004). It indicates the cost and availability of credit and savings services offered by financial institutions. A high IRS often reflects inefficiencies within the financial sector, such as elevated operational costs, limited competition, or weak regulatory frameworks, which may hinder the accessibility of financial services (Beck & Hesse, 2009; Demirgüç-Kunt & Klapper, 2012). Similarly, mobile cellular telephone subscriptions (MCS) are a critical indicator of financial inclusion in East African countries because they enable access to financial services without requiring a traditional bank account or internet connection. The widespread adoption of mobile money platforms such as Kenya's M-Pesa, Tanzania's Tigo Pesa, and Rwanda-Uganda's MTN Mobile Money has transformed how individuals save, borrow, and access other financial services(Demirgüç-Kunt et al., 2020; Mbiti & Weil, 2016). Mobile money services allow users to deposit and store money securely in digital wallets accessible via mobile phones. These platforms often collaborate with microfinance institutions and fintech companies to offer credit facilities. Since mobile money does not require internet access, it bridges the digital divide and extends financial services to rural and underserved populations(Mbiti & Weil, 2016).

Contrary to the previous studies that used GDP as an independent variable to measure economic growth, this study uses the Human Development Index (HDI) to measure economic development. HDI provides a multidimensional perspective by incorporating health (life expectancy), education (mean and expected years of schooling), and income (GNI per capita). In addition, this study employs Fully Modified Ordinary Least Squares (FMOLS) and Dynamic Ordinary Least Squares (DOLS), which are known to provide reliable long-run estimates in panel data models with cointegrated variables (Phillips & Hansen, 1990; Stock & Watson, 1993). A critical advantage of FMOLS and DOLS is that they inherently correct for and account for endogeneity, eliminate serial correlation, and adjust for heteroscedasticity (Kao & Chiang, 2000; Pedroni, 2001). Therefore, when using FMOLS and DOLS to estimate panel data models, diagnosing multicollinearity and cross-section dependence is crucial for ensuring the validity and reliability of the results(Kao & Chiang, 2000; Wooldridge, 2010).

This study suggests that financial inclusion indicators influence the Human Development Index (HDI) through multiple causal pathways. Four key indicators are examined: interest rate spread, mobile cellular subscriptions, gross domestic savings, and domestic bank credit to the private sector. These influences include (a) access to credit and

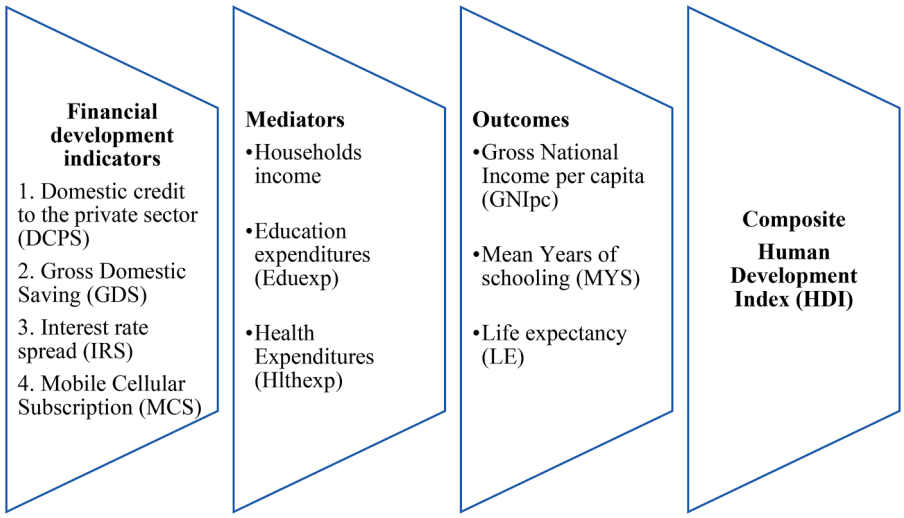

Fig. 1. From Financial Inclusion to HDI impact transmission mechanism.

investments, (b) savings and consumption smoothing, (c) ability to afford education affecting enrollment and years of schooling, and (d) capacity to finance healthcare and health insurance. Mobile subscriptions also enable digital access, improving educational, healthcare, and economic prospects. Collectively, these mediators impact the HDI's three sub-indices: income (via GNI per capita), education (through expenditure and years of schooling), and health (via expenditure and life expectancy).

Operationally, increased domestic credit to the private sector and a narrower interest rate spread are expected to reduce financing costs, thereby boosting productive investments and lifting household and firm incomes, which will positively impact the HDI income component. Higher gross domestic savings indicate a greater national capacity to fund investments and serve as buffers for households, enhancing income and the ability to afford education and healthcare expenses. An increase in mobile cellular subscription rates should improve access to mobile money and digital payments, reducing transaction costs and broadening the reach of financial services. This can lead to more school fee payments, increased use of health services, and support for income-generating activities. The, conceptual framework presented in Fig. 2 highlights this financial inclusion and HDI transmission mechanism. The mediation path model's estimation results are presented in the results section (Fig. 1).

3. Data and methodology

3.1. Variable selection

We measure financial inclusion using four indicators that reflect both market-level intermediation and digital access. First, the interest-rate spread (lending minus deposit rates) serves as a proxy for frictions in the financial sector intermediation: Large spreads indicate high borrowing costs and weak intermediation, which reduce access to affordable credit for households and firms, dampening investment, consumption smoothing, and ultimately slower improvements in HDI components, including income, capacity to pay for education, and health (Agapova & McNulty, 2016; Nuhiu & Aliu, 2024). Numerous policy studies link high spreads to reduced credit access and slower economic activity; central-bank and World Bank analyses discuss how spreads affect investment and welfare (Feyen, 2020). Spread is a market-level, supply-side metric. it does not directly measure individual account ownership or usage. It is best interpreted as an indicator of credit market depth and efficiency that conditions inclusion (Were & Wambua, 2014).

Second, the mobile cellular subscription rate proxies potential access to digital financial services, such as mobile money and payments. Empirical studies review demonstrate that mobile networks and mobile money are key channels for increasing financial inclusion and enhancing economic welfare (Jack, William, 2014). Mobile subscriptions are a proximate indicator of population access to mobile-based financial services such as mobile money, mobile banking (Aker et al., 2010). High mobile penetration increases the potential user base for digital payments, remittances, and other low-cost financial services that facilitate payments for education/health and improve market participation (Suri & Jack, 2016).

Third, gross domestic savings capture the economy's domestic capacity to finance investment and provide households with saving opportunities that smooth shocks and finance human capital expenditures (Grigoli et al., 2018). Savings have been linked in the growth and investment literature to improved economic outcomes (Schmidt-hebbel, 2014). Higher savings can support investments in health, education, and productive capital, which in turn affect HDI components. The literature linking savings to growth and investment is extensive (Loayza et al., 2014); empirical studies find that savings correlate with and Granger-cause GDP and investment in many contexts (Ribaj & Mexhuani, 2021). However, National savings are a macro aggregate and are not a direct measure of individual financial access. Therefore, it is used as a macro-level enabling condition because household savings data is not available.

Finally, domestic credit to the private sector is a standard measure of financial depth and the supply of credit available to private agents; it therefore relates directly to the system's ability to finance businesses and household investments that affect HDI components (Udoye et al., 2023). This indicator, which measures the credit to the private sector as a percentage of GDP, assesses the financial system's capacity and willingness to lend to households and firms. This is a standard proxy for the depth of financial intermediation and access to credit that supports business growth and household investment in education/health (Beck et al., 2000;Levine & Levine, 2004). We recognize that these indicators have limitations; for example, national aggregates can mask distributional differences.

3.2. Data sources and model description

This study uses econometric panel data from four East African Countries: Rwanda, Uganda, Kenya, and Tanzania. The dataset is based on yearly observations from 1999 to 2023, sourced from the World Bank Database for all independent variables and the United Nations Devel-

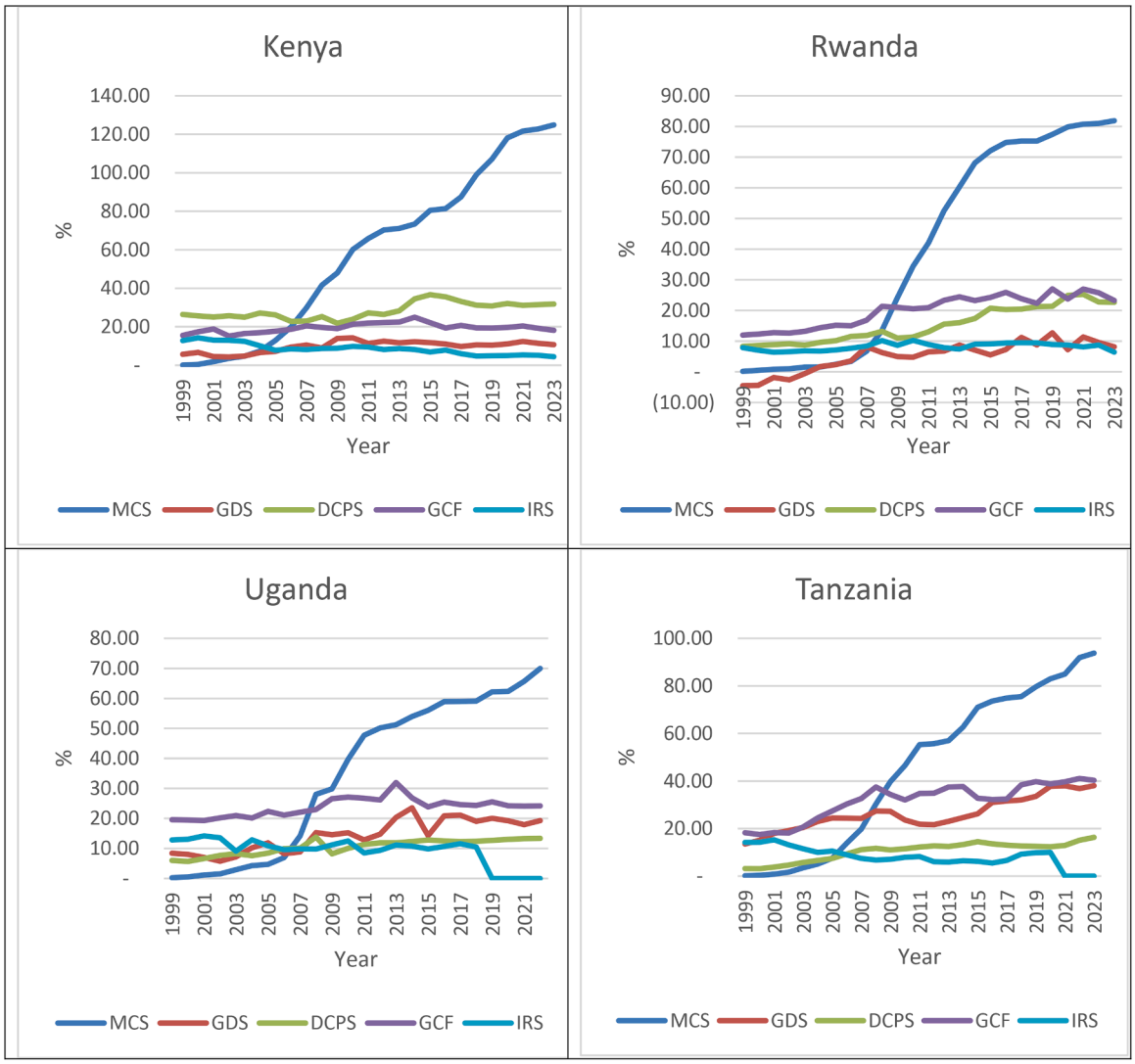

Fig. 2. The four-panel chart plots the percentages of five financial inclusion indicators for Kenya, Rwanda, Uganda, and Tanzania (1999–2023). It shows a rapid growth in MCS, starting near zero in the early 2000s and rising sharply from 2007 to 2009. By 2023, Kenya leads at over 120 %, followed by Tanzania (~95 %), Uganda (~75 %), and Rwanda (~85 %). Traditional channels show modest growth, with GDS (red) and IRS (light blue) slowly increasing but remaining below 20 % in all countries. DCPS (green) gradually rises to 20–30 % by 2023. Uganda's GCF (purple) peaks around 30 % in the mid-2010s before settling near 20 %. In Rwanda, non-MCS indicators are more compressed, indicating later or limited uptake of formal financial services outside mobile money. The synchronization of MCS acceleration reflects the region's mobile money rollout, but the pace and ceiling vary significantly by country. Traditional banking/service metrics (GDS, DCPS, IRS) remain flat across all four, highlighting how mobile platforms surpass legacy channels in driving inclusion.

opment Programme (UNDP) for the Human Development Index (HDI). This timeframe was selected due to the availability of data for all sampled countries. The dataset comprises variables representing financial inclusion, economic development represented by HDI, and the control variable Gross Capital Formation (GCF). The annual frequency was chosen to capture the potential long-term impact of financial inclusion on economic development, as significant changes in financial inclusion may take an entire year or more to manifest in changes to the HDI. Hence, Table 1 provides the names, abbreviations, measurements, and sources of the variables. The functional relationship between

Table 1
Variables, data measurement, and source.

Variable	Abbreviation	Measurement	Data source
Human Development Index	HDI	Between 1 and 10	https://hdr.undp.org/data-center/documentation-and-downloads
Interest Rate Spread	IRS	%	https://databank.worldbank.org/source/world-development-indicators#
Mobile cellular telephone subscription rate	MCS	per 100 people	https://databank.worldbank.org/source/world-development-indicators#
Gross Domestic Savings	GDS	% GDP	https://databank.worldbank.org/source/world-development-indicators#
Domestic credit to the private sector by banks	DCPS	% GDP	https://databank.worldbank.org/source/world-development-indicators#
Gross Capital Formation	GCF	% GDP	https://databank.worldbank.org/source/world-development-indicators#
Education Expenditure	Eduexp	% GDP	https://databank.worldbank.org/source/world-development-indicators#
Health Expenditure	Hlthexp	% GDP	https://databank.worldbank.org/source/world-development-indicators#
Gross National Income per capita	GNIpc	$	https://databank.worldbank.org/source/world-development-indicators#

dependent and independent variables is expressed in the Eq. (1) as follows:

$$\boldsymbol{lnHDI_{it}} = \beta_0 + \beta_1 \ln \boldsymbol{MCS_{it}} + \beta_2 \ln \boldsymbol{IRS_{it}} + \beta_3 \boldsymbol{lnGDS_{it}} + \beta_4 \boldsymbol{lnDCPS_{it}} + \beta_5 \boldsymbol{lnGCF_{it}} + \boldsymbol{\alpha_i} + \boldsymbol{\mu_{it}} \quad (1)$$

Where:

Ln: The natural logarithm; "i" represents a country; "t" the time or the year; "α_i" the Individual Effects; "μ_{it}" the Error Term, and "β" the Coefficients to be estimated.

Fig. 2. Visually represents the temporal trends in the study variables across the sampled countries. It allows for a comparative analysis of their evolution over time, highlighting potential similarities, differences, and underlying patterns.

Fig. 3 provides a comparative analysis of study variables across sampled countries from 1999 to 2023. This figure visually examines the trends and patterns in each country and the overall sample.

3.3. Methodology

This paper utilizes a four-stage approach, including panel unit root tests, panel cointegration tests, estimation of panel coefficients, robustness and diagnostic tests. Following stationarity and cointegration analysis and the estimation of long-run relationships, we implemented empirical analysis to estimate the effects of financial indicators on mediators. We also performed Granger-causality tests as part of our robustness and diagnostic stage. Specifically, we apply panel Granger-causality procedures to assess the direction of predictive influence between each HDI subindex (income, health, and education) and its proposed mediators (income, health expenditure, and education expenditure). These causality checks complement the main coefficient estimates by testing temporal precedence and helping to distinguish predictive directionality from mere association. The results are reported in the results section and used to strengthen the causal interpretation.

3.3.1. Panel unit root tests

The variables used in panel data analysis should be non-stationary at their level form, but become stationary after taking the first difference (Mburamatare et al., 2022b). This ensures that potential spurious relationships among the variables are avoided. Two categories of tests are employed to evaluate the stationarity properties of the variables in this study. The first is the common unit root process, which includes the Levin, Lin & Chu (LLC) test. This test assumes that all cross-sections (countries) share a common unit root process, meaning they have a similar degree of persistence (Levin et al., 2002). The second category is the individual unit root process, including the Im, Pesaran, and Shin (IPS) W-stat, ADF-Fisher, and PP-Fisher Chi-square tests. This second category allows for heterogeneity in the unit root process across cross-sections, meaning countries can have varying levels of persistence (Im et al., 2003; Maddala & Wu, 1999; Phillips & Perron, 1988).

The Eq. (2) represents the LLC test. This test is particularly robust when cross-sections have similar dynamics. However, it assumes a common ρ, which might not always hold in heterogeneous panels (Levin et al., 2002). Alternative tests, like IPS represented by the Eq. (3) or Fisher-based tests represented by the Eqs. (4) and (5) are more suitable when heterogeneity across cross-sections is expected. The representative equation for the LLC test is as follows:

$$\Delta \boldsymbol{y_{it}} = \boldsymbol{\alpha_i} + \boldsymbol{\rho_i y_{i,t-1}} + \sum_{j=1}^{p} \phi_{ij} \Delta \boldsymbol{y_{i,t-j}} + \boldsymbol{\varepsilon_{i,t}};\ \boldsymbol{i} = 1, 2, \ldots, \boldsymbol{N};\ \boldsymbol{t} = 1, 2, \ldots, \boldsymbol{T} \quad (2)$$

where y_{it} stands for each variable under consideration in our model, α_i is the individual fixed effect, $\varepsilon_{i,t}$ is the error term, ϕ_{ij} represents the coefficients of lag differenced of variables and ρ_i is selected to make the residuals uncorrelated over time.

The representative equation for the IPS test is as follows:

$$\Delta \boldsymbol{y_{it}} = \boldsymbol{\rho_i y_{i,t-1}} + \sum_{j=1}^{p} \phi_{ij} \Delta \boldsymbol{y_{i,t-j}} + \boldsymbol{\varepsilon_{i,t}};\ \boldsymbol{i} = 1, 2, \ldots, \boldsymbol{N};\ \boldsymbol{t} = 1, 2, \ldots, \boldsymbol{T} \quad (3)$$

The representative equation for the ADF - Fisher Chi-square test is as follows:

$$\Delta \boldsymbol{y_{it}} = \boldsymbol{\alpha_i} + \boldsymbol{\beta_i t} + \boldsymbol{\rho_i y_{i,t-1}} + \sum_{j=1}^{p} \phi_{ij} \Delta \boldsymbol{y_{i,t-j}} + \boldsymbol{\varepsilon_{i,t}};\ \boldsymbol{i} = 1, 2, \ldots, \boldsymbol{N};\ \boldsymbol{t} = 1, 2, \ldots, \boldsymbol{T} \quad (4)$$

Like the ADF-Fisher test, the PP-Fisher test accommodates heterogeneity across cross-sections, but it uses a nonparametric method to correct for serial correlation and heteroscedasticity in the error terms (Phillips & Perron, 1988). The representative equation of the Phillips-Perron (PP) - Fisher Chi-square test is as follows:

$$\boldsymbol{y_{it}} = \boldsymbol{\alpha_i} + \boldsymbol{\beta_i t} + \boldsymbol{\rho_i y_{i,t-1}} + \boldsymbol{\varepsilon_{i,t}};\ \boldsymbol{i} = 1, 2, \ldots, \boldsymbol{N};\ \boldsymbol{t} = 1, 2, \ldots, \boldsymbol{T} \quad (5)$$

Therefore, to ensure the model's robustness, all the presented panel unit root tests are conducted and their results are presented. In addition, once all variables are stationary with the same order of integration, we can conduct co-integration tests to determine whether a long-run relationship or equilibrium exists among the variables(Daniel et al., 2023; Mburamatare et al., 2022b).

3.3.2. Panel cointegration test

Panel cointegration tests assess whether a long-run equilibrium relationship exists between a dependent variable and a set of independent variables in panel data. To ensure the reliability of the results, this study uses three cointegration tests, including Pedroni, Kao, and Johansen Fisher Panel Cointegration Test. The three tests differ in methodology, assumptions, and implementation. The Pedroni Panel Cointegration test extends Engle-Granger's residual-based approach to panel data and allows for heterogeneity in cointegration vectors and individual-specific dynamics(Pedroni, 1999, 2004). The Pedroni Panel Cointegration Test model is specified in the Eq. (6) as follows:

$$\boldsymbol{y_{it}} = \boldsymbol{\alpha_i} + \boldsymbol{\delta_i t} + \boldsymbol{\beta_i x_{it}} + \boldsymbol{e_{i,t}}; \quad (6)$$

Where:

- y_{it}: Dependent variable (HDI),
- x_{it}: Independent variables (MCS, IRS, DCPS, GDS, and GCF),
- α_i: Individual-specific intercept,
- δ_{it}: Time trend,
- β_i: Heterogeneous slope coefficients,
- e_{it}: Residuals.

The Kao Residual Cointegration Test is similar to Pedroni but assumes homogeneity in the cointegrating relationship across cross-sections(Kao, 1999). The Kao Residual Cointegration Test model is represented in the Eq. (7) as follows:

$$\boldsymbol{y_{it}} = \boldsymbol{\alpha_i} + \boldsymbol{\beta_i x_{it}} + \boldsymbol{e_{i,t}} \quad (7)$$

And the residuals $(\boldsymbol{e}(\boldsymbol{i}, \boldsymbol{t}))$ follow $\boldsymbol{e_{i,t}} = \boldsymbol{\rho e_{i,t-1}} + \boldsymbol{\mu_{it}}$

Where ρ measures the persistence of residuals.

Lastly, the Johansen-Fisher Panel Cointegration Test extends the Johansen cointegration test for individual time series to panel data by combining individual test results using Fisher's aggregation method (Maddala & Wu, 1999). This model is estimated as a Vector Error Correction Model (VECM) as represented in the Eq. (8) and uses fisher's method aggregates test results as represented in Eq. (9) as follows:

$$\Delta \boldsymbol{Y_{it}} = \prod \boldsymbol{Y_{i,t-1}} + \sum_{k=1}^{p-1} \Gamma_k \Delta \boldsymbol{Y_{i,t-k}} + \varepsilon_{i,t} \quad (8)$$

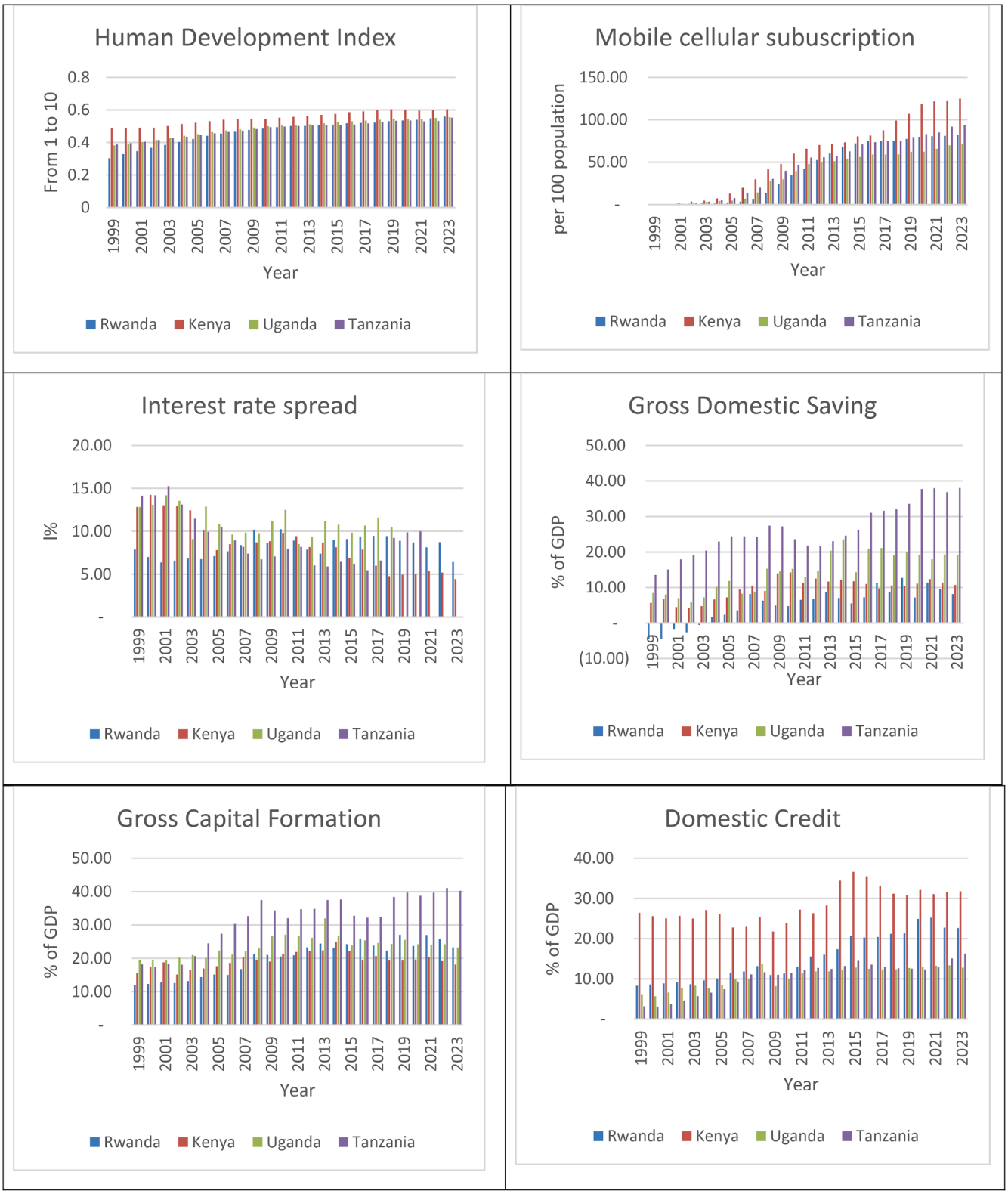

Fig. 3. The six-panel chart compares four East African countries (Rwanda, Kenya, Uganda, Tanzania) on six key variables from 1999 to 2023. All four countries exhibit a steady upward trend in HDI, with Kenya and Tanzania consistently ranking above Rwanda and Uganda. Mobile Cellular Subscriptions, near-zero in 1999, exploded after 2007. Kenya leads, followed by Tanzania, Uganda, and Rwanda. The interest rate spread remains high (roughly 8–18 percentage points) and volatile in each country, with no clear downward trend or convergence. Gross Domestic Saving gradually rises for all countries, with Kenya and Tanzania moving from the low teens toward the high 20 s, while Rwanda and Uganda lag slightly. Gross Capital Formation increases across the board; Tanzania shows the strongest growth, Kenya and Uganda experience moderate gains, and Rwanda records the slowest uptick. Domestic Credit to the Private Sector grows unevenly: Kenya peaks above 80 % of GDP around 2015 before easing back, while Tanzania and Uganda show moderate, steady increases, and Rwanda remains the most conservative.

Where:

- Y_{it}: The vector of dependent variable (HDI) and independent variables,
- $\Pi = \alpha\beta'$: The cointegration matrix,
- β: Cointegrating vectors,
- α_i: adjustment parameters,

$$\chi 2 = -2\sum_{i=1}^{N} \ln(p_{i,}) \tag{9}$$

Where p_i are the p-values of the individual Johansen test.

3.3.3. The long-run relationship estimation models

The study utilizes a Panel Fully Modified Least Squares (FMOLS) and Dynamic Ordinary Least Squares (DOLS) to estimate the model coefficients as expressed in Eq. (1). This equation highlights the relationship between dependent and independent variables. The FMOLS and DOLS methods offer complementary insights into the long-run relationships between HDI and the independent variables. They provide reliable long-run estimates in panel data models with cointegrated variables. While FMOLS modifies OLS estimates using covariance corrections (Pedroni, 2001), DOLS enhances robustness by incorporating dynamics through leads and lags (Stock & Watson, 1993). These methods outperform OLS when dealing with endogeneity, autocorrelation, and non-stationarity (Kao & Chiang, 2000). The FMOLS modifies the Ordinary Least Squares (OLS) estimator to account for endogeneity and serial correlation in the residuals, while the DOLS extends the model by incorporating leads and lags of first-differenced independent variables to control for endogeneity(Baltagi, 2005; Pedroni, 2001; Stock & Watson, 1993).

The Eq. (10) represents the FMOLS model as follows:

$$Y_{it} = \beta_0 + \beta_1 X_{1,\ it} + \beta_2 X_{2,\ it} + \ldots + \beta_k X_{k,\ it} + u_{it} \tag{10}$$

Where:

- y_{it}: Dependent variable (HDI),
- x_{it}: Independent variables (MCS, IRS, DCPS, GDS, and GCF),
- u_{it}: Error term, assumed to have serial correlation and potential endogeneity with $X_{k,\ it}$.

The Eq. (11) represents the FMOLS estimator as follows:

$$\widehat{\beta}_{FMOLS} = \left(\sum_{i=1}^{N}\sum_{t=1}^{T} \widetilde{X}_{i,t}\widetilde{X}'_{i,t}\right)^{-1} \left(\sum_{i=1}^{N}\sum_{t=1}^{T} \widetilde{X}_{i,t}\widetilde{Y}_{i,t}\right) \tag{11}$$

where:

- $\widetilde{X}_{i,t}$: Transformed independent variables after correcting for endogeneity.
- $\widetilde{Y}_{i,t}$: Adjusted dependent variable.

The DOLS model is represented by the Eq. (12) as follows:

$$Y_{it} = \alpha_i + X'_{it}\beta + \sum_{j=-q_1}^{j=q_2} c_{ij}\Delta X_{i,t+j} + \mu_{it} \tag{12}$$

where:

Y_{it}: represents the dependent variable,
X: Explanatory variables,
β: denotes the coefficients of explanatory variables,
c_{ij}: represents the coefficients of lag-differenced variables,
α_i: Individual Effects and μ_{it} denotes the error term.
The Eq. (13) represents the DOLS estimator as follows:

$$\widehat{\beta}_{DOLS} = \left(\sum_{i=1}^{N}\sum_{t=1}^{T} Z'_{it}Z_{it}\right)^{-1} \left(\sum_{i=1}^{N}\sum_{t=1}^{T} Z'_{it}Y_{it}\right) \tag{13}$$

Where $Z_{it} = [X_{\ it}, \Delta X_{i,t-j}, \ldots, \Delta X_{i,t+j}]$ includes the original regressors and their differences.

3.3.4. Diagnostic tests

When using FMOLS and DOLS to estimate panel data models, diagnosing multicollinearity and cross-section dependence is crucial for ensuring the validity and reliability of the results. While multicollinearity inflates standard errors, leading to unreliable estimates and incorrect inferences about the significance of explanatory variables (Gujarati & Porter, 2009), ignoring cross-section dependence can lead to biased and inconsistent estimates(Pesaran, 2004). However, FMOLS corrects endogeneity by applying a non-parametric adjustment to the residuals based on the covariance of the regressors and the error term (Kao, 1999). DOLS accounts for endogeneity by including leads and lags of the differenced regressors to remove the correlation between the regressors and the error term(Baltagi, 2005). In addition, FMOLS uses kernel-based corrections to the residuals' covariance matrix to eliminate serial correlation, and DOLS eliminates serial correlation by modeling dynamic adjustments using different terms. Lastly, FMOLS adjusts for heteroscedasticity through robust variance-covariance estimation of residuals, and DOLS produces heteroscedasticity-robust standard errors, ensuring valid statistical inference(Baltagi, 2005; Pedroni, 1999, 2004). Therefore, for diagnosing multicollinearity, this study uses Variance Inflation Factors (VIFs) calculated through the Eq. (14) as follows:

$$VIF_j = \frac{1}{1 - R_j^2} \tag{14}$$

R_j^2: The coefficient of determination (R-squared) from the regression of the jth independent variable on all other independent variables.

Eq. (14) indicates that VIF_j depends on R_j^2, This means that as the correlation among independent variables increases, the VIF also rises (Mburamatare et al., 2022a, 2022b). The goal is to achieve lower VIF values. A threshold of 10 is commonly used, with values below 10 indicating no significant multicollinearity and above 10 suggesting that multicollinearity poses a challenge for estimating coefficients. This study uses various tests concerning cross-sectional dependence diagnostics, including Pesaran's Cross-Section Dependence (CD), Pesaran scaled LM, Bias-corrected scaled LM, and Breusch-Pagan LM Test. Pesaran's CD test statistic is computed in the Eq. (15) as follows:

$$CD = \sqrt{\frac{2T}{N(N-1)}} \sum_{i=1}^{N-1}\sum_{j=i+1}^{N} \widehat{\rho}_{ij} \tag{15}$$

Where:

- **T**: Time dimension (number of time periods); **N**: Cross-sectional units (e.g., countries).
- $\hat{\rho}_{ij}$: Pairwise correlation of residuals between cross-sections **i** and **j**.

The Breusch-Pagan LM Test for cross-sectional dependence is computed in Eq. (16) as follows:

$$LM = \sum_{i=1}^{N-1}\sum_{j=i+1}^{N} T\widehat{\rho}_{ij}^2 \tag{16}$$

Where:

- **T** is the time dimension (number of periods); $\hat{\rho}_{ij}^2$ is Squared correlational residuals.

4. Results and discussion

This section presents and discusses the study's findings. First, we present the results of the panel unit root test. Then, we present the Panel cointegration test results. Lastly, we present and discuss the models' estimation, robustness, and Diagnostic Tests.

4.1. Panel unit root test results

The panel unit root test has been conducted to establish whether the variables are stationary or non-stationary at "level" and in "first differences". Their statistics are reported in Tables 2 and 3. At level, most variables exhibit non-stationarity across the majority of tests. Only HDI and GCF show signs of stationarity under specific tests (LLC and PP-Fisher), but the results are inconsistent across all methods.

Therefore, further testing at the first difference has been conducted to assess stationarity for all variables. All tests at the first difference reject the null hypothesis at a 1 % significance level for IRS, GDS, DCPS, and GCF variables. Regarding MCS, all tests reject the null hypothesis at a p-value of less than 3 %, while for HDI, the null hypothesis is rejected at a 5 % significance level. This indicates that the non-stationarity present at level form has been resolved. Therefore, we can conclude that all variables are non-stationary and integrated with order one, denoted as I (1). Therefore, since all variables are stationary at the first difference and integrated of the same order (I (1)), we can proceed with co-integration tests to assess whether a long-run relationship or equilibrium exists among the variables; this corroborates the results found by Daniel et al. (2023).

4.2. Cointegration tests

With all variables confirmed to be integrated of the same order, the Pedroni, Kao, and Johansen Fisher co-integration tests are employed to assess the presence of a long-run relationship or equilibrium among the variables in the model. The Pedroni Residual Cointegration Test results presented in Table 4 provide strong evidence of a long-term equilibrium relationship among the variables in the study. Specifically, the Panel PP-Statistic and Panel ADF-Statistic within-dimension tests, along with the Group PP-Statistic and Group ADF-Statistic between-dimension tests, reject the null hypothesis of no cointegration ($p < 0.05$), indicating the presence of cointegration. Conversely, the Panel v-statistic, Panel rho-statistic, and Group rho-statistic fail to reject the null hypothesis ($p > 0.05$), suggesting mixed results across metrics. The results indicate that six out of eleven tests reject the null hypothesis at a 1 % significance level. Since most tests support rejecting the null hypothesis, it can be concluded that the model's variables are cointegrated, indicating a long-run relationship.

However, to ensure the robustness of Pedroni's test results, the Kao and Johansen Fisher co-integration tests are conducted to validate these initial findings. Kao's residual cointegration test results are presented in Table 5, and Johansen Fisher Panel cointegration test results in Table 6. The Kao Residual Cointegration Test results confirm a long-term equilibrium relationship among the variables in the study, providing robust support for the earlier Pedroni test results. The test rejects the null hypothesis of no cointegration, with a highly significant ADF t-statistic ($p < 0.01$) and evidence of stationary and mean-reverting residuals. Model diagnostics indicate reliability and a good fit, with no significant autocorrelation.

The results in Table 6 indicate at least one cointegration vector with strong evidence from Trace and Max-Eigen tests (p-value = 0.0000). The tests also show evidence for up to three cointegration vectors with consistent results, suggesting that three is the most likely number of long-term relationships. Weaker evidence is found for four and five cointegration vectors, with p-values above 5 %, indicating less confidence in these relationships. In conclusion, the test validates the Pedroni and Kao Residual Cointegration Tests, confirming the existence of long-term equilibrium relationships among the independent variables. Hence, these results validate the use of advanced econometric methods, such as FMOLS or DOLS, to accurately estimate the long-run relationships between the variables.

4.3. Models' estimation results

The results of the FMOLS, as presented in Table 7 and Eq. (17), (17) reveal that mobile cellular subscriptions, gross domestic savings, and domestic credit to the private sector have significant and positive effects on economic development, as represented by HDI. Specifically, a 1 % increase in these variables leads to increases of 0.05 %, 0.09 %, and 0.05 % in HDI, respectively, underscoring their critical role in fostering development. This corroborates various studies' findings in one way or another(Fuseini et al., 2024; Gizaw, Getachew & Mancha, 2024a, 2024b). Conversely, the interest rate spread and gross capital formation have a negative impact on the HDI, with coefficients of −0.065 and −0.34, respectively. These findings suggest that higher borrowing costs and potential inefficiencies in capital allocation may hinder progress in human development. In conclusion, the FMOLS analysis underscores the importance of enhancing access to financial services, promoting savings,

Table 2
Panel unit root test results of the variables at level.

Null Hypothesis: Has a Unit Root → non-stationary
Alternate Hypothesis: Does not have Unit Root →Stationary

Methods		Levin, Lin & Chu (LLC)		Im, Pesaran and Shin (IPS) W-stat		ADF-Fisher Chi-square		PP-Fisher Chi-square	
		Constant	Constant & trend	Constant	Constant & trend	Constant	Constant & trend	Constant	Constant & trend
Variables									
Level	**HDI**	−3.73795	−1.93292	−1.07725	−0.00496	12.7097	6.27651	45.5280	6.22445
	Prob	(0.0001)	(0.0266)	(0.1407)	(0.4980)	(0.1222)	(0.6163)	(0.0000)	(0.6221)
	MCS	−1.41948	−0.65733	1.28386	−0.60371	2.86604	9.81228	0.64219	4.80797
	Prob	(0.0779)	(0.2555)	(0.9004)	(0.2730)	(0.9425)	(0.2784)	(0.9997)	(0.7779)
	IRS	−1.43115	0.41780	−1.12688	−0.19480	12.2886	13.8754	11.2312	8.29925
	Prob	(0.0762)	(0.6620)	(0.1299)	(0.4228)	(0.1388)	(0.0851)	(0.1889)	(0.4048)
	GDS	−1.43095	−1.48056	−0.23910	−1.36076	7.20277	12.6212	5.97938	9.51129
	Prob	(0.0762)	(0.0694)	(0.4055)	(0.0868)	(0.5149)	(0.1256)	(0.6495)	(0.3010)
	DCPS	−1.26708	−1.68139	0.11213	−1.53489	5.82693	13.6398	4.20199	10.9117
	Prob	(0.1026)	(0.0463)	(0.5446)	(0.0624)	(0.6666)	(0.0917)	(0.8385)	(0.2068)
	GCF	−1.82205	−1.15517	−0.66350	0.76991	8.40846	3.48969	8.09107	2.89593
	Prob	(0.0342)	(0.1240)	(0.2535)	(0.7793)	(0.3946)	(0.9000)	(0.4246)	(0.9407)

Notes: p-values are in parentheses, and the significance level is $p < 0.05$.

Table 3
Panel unit root test results of the variables at first difference.

Null Hypothesis: Has a Unit Root → non-stationary Alternate Hypothesis: Does not have Unit Root →Stationary									
Methods		Levin, Lin & Chu (LLC)		Im, Pesaran and Shin (IPS) W-stat		ADF-Fisher Chi-square		PP-Fisher Chi-square	
		Constant	Constant & trend	Constant	Constant & trend	Constant	Constant & trend	Constant	Constant & trend
Variables									
First difference	**ΔHDI**	−1.96563	−1.62607	−2.02381	−1.70938	17.7651	20.2784	13.8295	10.4490
	Prob	(0.0247)	(0.0520)	(0.0215)	(0.0437)	(0.0231)	(0.0093)	(0.0863)	(0.2349)
	ΔMCS	−1.81241	−2.61196	2.61299	–5.71215	21.2671	17.9425	19.3351	16.7783
	Prob	(0.0350)	(0.0045)	(0.0045)	(0.0000)	(0.0065)	(0.0217)	(0.0132)	(0.0325)
	ΔIRS	−5.67008	−7.43090	−5.52356	−7.69518	49.5163	54.8177	72.8615	64.5599
	Prob	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
	ΔGDS	−7.87945	−4.21585	−9.07071	−6.58566	72.3723	49.6126	87.8237	126.223
	Prob	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
	ΔDCPS	−7.51492	−3.09089	−6.19364	−2.95674	49.3125	23.1358	69.5018	283.547
	Prob	(0.0000)	(0.0010)	(0.0028)	(0.0016)	(0.0071)	(0.0032)	(0.0000)	(0.0000)
	ΔGCF	−8.35474	−5.83783	−7.62135	−5.81289	59.8746	42.2699	70.1602	154.192
	Prob	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)

Notes: p-values are in parentheses, and the significance level is $p < 0.05$.

Table 4
Pedroni panel cointegration test results.

Pedroni Residual Cointegration Test					
Methods	Within dimension (Panel statistics)			Between dimension (Individuals statistics)	
	Test	**Statistics**	**Prob.**	**Statistics**	**Prob.**
Pedroni	Panel v-statistic	−0.048410	0.5193	–	–
	Panel rho-statistic	0.611344	0.7295	1.117602	0.8681
	Panel PP-statistic	−2.358816	0.0092	−2.621133	0.0044
	Panel ADF-statistic	−2.369122	0.0089	2.678787	0.0037
Pedroni	Panel v-statistic	−0.224810	0.5889	–	–
(Weighted statistic)	Panel rho-statistic	0.323391	0.6268	–	–
	Panel PP-statistic	−2.223511	0.0131	–	–
	Panel ADF-statistic	−2.214061	0.0134	–	–

Notes: The significance level is $p < 0.05$.

Table 5
Kao residual cointegration test results.

Null Hypothesis: No Cointegration		
ADF	t-Statistic	Prob.
	−6.417181	0.0000
Residual Variance	0.089256	
HAC Variance	0.167345	

Table 6
Johansen Fisher Panel cointegration test results.

Unrestricted cointegration rank test (Trace and maximum eigenvalue)				
Hypothesized No. of CE(s)	Fisher Stat.* (From trace test)	Prob.	Fisher Stat.* (From max-eigen test)	Prob.
None	238.1	0.0000	134.2	0.0000
At most 1	137.7	0.0000	92.95	0.0000
At most 2	65.24	0.0000	34.20	0.0000
At most 3	37.67	0.0000	29.70	0.0002
At most 4	16.17	0.0400	14.34	0.0734
At most 5	13.36	0.1001	13.36	0.1001

* Probabilities are computed using the asymptotic Chi-square distribution.

Table 7
Panel FMOLS long-run estimation results.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LNMCS	0.050602	0.007113	7.114450	0.0000
LNIRS	−0.064655	0.025183	−2.567440	0.0122
LNGDS	0.087767	0.015934	5.508034	0.0000
LNDCPS	0.054203	0.013491	4.017681	0.0001
LNGCF	−0.337398	0.028536	−11.82352	0.0000
R-squared	0.828494	Mean dependent var		−0.687754
Adjusted R-squared	0.819698	S.D. dependent var		0.105482
S.E. of regression	0.044790	Sum squared resid		0.156476
Long-run variance	0.002651			

Dependent Variable: LNHDI.
Method: Panel Fully Modified Least Squares (FMOLS).
Sample (adjusted): 2000 2023; Periods included: 24 and Cross-sections included: 4.
Total panel (unbalanced) observations: 83.
Panel method: Pooled estimation

and addressing inefficiencies in capital investments to promote human development in East Africa.

4.3.1. The FMOLS model's representative equation

$$\boldsymbol{Log}\ (\boldsymbol{HDI}) = 0.05*\boldsymbol{log}\ (\boldsymbol{MCS}) - 0.06*\boldsymbol{log}\ (\boldsymbol{IRS}) + 0.09*\boldsymbol{log}\ (\boldsymbol{GDS}) + 0.05*\boldsymbol{log}\ (\boldsymbol{DCPS}) - 0.34*\boldsymbol{log}\ (\boldsymbol{GCF}) \tag{17}$$

In addition, this study also employs the Panel Dynamic Least Squares (DOLS) methodology to validate the results from FMOLS model estimation. The findings presented in Table 8 and Eq. (18) reveal that mobile cellular subscriptions (0.04 %), gross domestic savings (0.02 %), and domestic credit to the private sector (0.08 %) positively and significantly contribute to economic development represented by HDI, as evidenced by their coefficients and low p-values. These results align with several studies' findings (Atta-Ankomah, Adjei-Mantey & Amankwah, 2024; Fuseini et al., 2024; He & Yoo, 2024). Conversely, the interest rate spread (−0.05 %) and gross capital formation (−0.14 %) negatively impact HDI. The latter, while counterintuitive, suggests inefficiencies in investment quality or allocation that merit further investigation. All variables are statistically significant at the 5 % level. The model demonstrates exceptional explanatory power, with an R-squared of 99 % and an adjusted R-squared of 99 %, confirming its ability to capture the variation in HDI effectively. A low standard regression error (0.003) highlights the model's precision.

Table 8
Panel DOLS long-run estimation results.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LNMCS	0.041873	0.003038	13.78093	0.0000
LNIRS	−0.055008	0.008913	−6.171849	0.0008
LNGDS	0.025699	0.009666	2.658739	0.0376
LNDCPS	0.083909	0.011230	7.471927	0.0003
LNGCF	−0.141328	0.021551	−6.557848	0.0006
R-squared	0.999914	Mean dependent var		−0.684091
Adjusted R-squared	0.998933	S.D. dependent var		0.097703
S.E. of regression	0.003191	Sum squared resid		6.11E-05
Long-run variance	2.70E-07			

Dependent Variable: LNHDI.
Method: Panel Dynamic Least Squares (DOLS).
Sample (adjusted): 2001 2022; Periods included: 22 and Cross-sections included: 4.
Total panel (unbalanced) observations: 75.

4.3.2. The DOLS model representative equation

$$\boldsymbol{Log}\ (\boldsymbol{HDI}) = 0.04*\boldsymbol{log}\ (\boldsymbol{MCS}) - 0.05*\boldsymbol{log}\ (\boldsymbol{IRS}) + 0.03*\boldsymbol{log}\ (\boldsymbol{GDS}) + 0.08*\boldsymbol{log}\ (\boldsymbol{DCPS}) - 0.14*\boldsymbol{log}\ (\boldsymbol{GCF}) \tag{18}$$

Therefore, both FMOLS and DOLS models highlight the positive and significant impacts of MCS, GDS, and DCPS on HDI. The magnitude of these effects varies slightly between the models. FMOLS estimates highlight more substantial positive impacts for GDS (0.09 %) and DCPS (0.05 %) compared to DOLS (0.03 % and 0.08 %, respectively). Conversely, both models agree on the negative impacts of IRS and GCF on HDI. However, the negative effect of GCF is much more substantial in FMOLS (−0.34) than in DOLS (−0.14). This discrepancy may reflect methodological differences in handling autocorrelation and endogeneity. Both methods emphasize the importance of access to mobile technology, savings, and credit availability, while highlighting the detrimental effects of inefficient capital allocation and high borrowing costs.

The negative correlation between Gross Capital Formation (GCF) and the Human Development Index (HDI) in East African countries may be attributed to various economic and structural factors. Investment misallocation can hinder economic development, as funds allocated to low-productivity projects limit the socio-economic benefits. Prioritizing physical infrastructure over crucial social spending in health and education undermines HDI improvements. Additionally, Debt-Financed Investment often relies on external debt, diverting resources from social sectors for short-term gains instead of long-term human development. Lagged effects explain the negative GCF sign in the model, as capital investments, such as infrastructure, take time to impact health and education, delaying HDI improvements. Lastly, structural economic challenges impact capital-intensive investments in low-wage economies, restricting technological advancements and hindering equitable growth and productivity.

4.4. Robustness tests

4.4.1. Mediation path models/ mediator regressions

Table 9 reports the estimated effects of financial-inclusion indicators on the mediators, estimated using panel FMOLS.

For the income model (lnGNI), domestic credit to the private sector (lnDCPS), gross domestic savings (lnGDS), and gross capital formation (lnGCF) have a strong, positive, and highly significant effect on the income component of the HDI. In contrast, mobile cellular subscriptions (lnMCS) and interest-rate spread (lnIRS) are not statistically significant. In the education model (lnEduexp), only lnDCPS shows a significant positive relationship; the other variables are statistically insignificant, indicating limited evidence that these financial indicators have a long-term impact on public education spending. Considering the health model (lnHlthexp), lnDCPS again has a strong positive effect, while lnIRS is negative and significant, suggesting that wider interest-rate spreads are associated with lower health expenditure. Other coefficients are not significant. Overall, the dominant and consistent finding is that greater private-sector credit, along with higher gross capital formation and savings, are the key long-run correlates of higher GNI and increased public spending on education and health. Mobile penetration shows no robust long-run effect here. These results highlight that some financial inclusion indicators indirectly affect HDI through proposed mediators, while others affect it directly without going through mediators.

4.4.2. Measuring the relative contribution of HDI components

The results in Table 10 show that the impact of individual HDI components on the overall HDI index varies greatly. Per capita national income is the most influential and significant factor, with a highly positive coefficient of 0.400, indicating it is the main driver of HDI in this model. Health expenditure also exhibits a statistically significant positive correlation with HDI, although its coefficient (0.049) indicates a smaller marginal effect compared to income. Interestingly, education expenditure appears to have no statistically significant impact on HDI. Although the model fits the data very well (R-squared = 0.953), it suggests that within this dataset, income and health spending are the primary measurable factors influencing HDI. In contrast, the effect of education spending is minimal.

4.4.3. Assessment of potential reverse causality

As presented in Table 11, the investigation employed Pairwise Granger Causality Tests, revealing that specific aspects of financial inclusion drive human development. Specifically, Domestic Credit to the

Table 10
HDI components relative contributions.

Dependent Variable: LNHDI				
Method: Panel FMOLS				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
LNHLTHEXP	0.04899	0.01731	2830,670	0.0059
LNGNIPC	0.40017	0.02713	14.74844	0.0000
LNEDUEXP	0.00048	0.02471	0.019588	0.9844

Table 9
Mediation path models estimation results.

Independent Variables	**Income model**		**Education Model**		**Health Model**	
	Dependent: lnGNI		Dependent: lnEduexp		Dependent: lnHlthexp	
	Coefficient	Prob.	Coefficient	Prob.	Coefficient	Prob.
lnMCS	−0.14	0.17	−0.10	0.34	0.18	0.13
lnIRS	−0.13	0.23	−0.006	0.95	−0.32	0.01
lnGDS	0.28	0.04	0.046	0.72	0.23	0.13
lnGCF	1.39	0.00	−0.005	0.97	0.06	0.74
lnDCPS	1.27	0.00	0.54	0.00	1.02	0.00

Table 11
Granger causality test results.

Null Hypothesis:	Obs	F-Statistic	Prob.
LNMCS does not Granger Cause LNHDI	92	0.19427	0.8238
LNHDI does not Granger Cause LNMCS		7.30291	0.0012
LNGDS does not Granger Cause LNHDI	87	2.08718	0.1306
LNHDI does not Granger Cause LNGDS		0.63725	0.5313
LNDCPS does not Granger Cause LNHDI	92	4.10572	0.0198
LNHDI does not Granger Cause LNDCPS		0.13673	0.8724
LNGCF does not Granger Cause LNHDI	92	3.31045	0.0412
LNHDI does not Granger Cause LNGCF		1.95299	0.1480
LNIRS does not Granger Cause LNHDI	84	1.89467	0.1571
LNHDI does not Granger Cause LNIRS		0.58245	0.5609

Private Sector and Gross Capital Formation Granger-cause HDI, with p-values of 0.0198 and 0.0412, indicating that past private sector credit and physical investment help predict current HDI levels. Conversely, a strong reverse causal link suggests that improvements in HDI lead to increased Mobile Cellular Subscriptions (LNMCS), indicating that healthier, educated, and wealthier populations are more likely to adopt digital financial technologies.

Even if FMOLS and DOLS correct for endogeneity, to go beyond correlation and estimate the causal impact of financial inclusion on HDI while controlling for endogeneity and unobserved heterogeneity, Two-Stage Least Squares (2SLS) and Generalized Method of Moments (GMM) estimators were used. The estimates in Table 12 indicate that all financial inclusion indicators exhibit significant elasticities with respect to HDI. The J-statistic p-values are significant, so we accept the null of instrument exogeneity, the instruments appear to be appropriate, and the coefficients are meaningful and robust. GMM and 2SLS estimates match because GMM generalizes IV and reduces to 2SLS under the same conditions, yielding identical estimates and J-test results.

4.4.4. Country-level regressions

Table 13 reports Country-level time-series results. The country regressions reveal substantial heterogeneity in how financial-sector and macroeconomic variables affect HDI over time. Mobile subscriptions are positively and significantly associated with HDI in Rwanda, Uganda, and Tanzania, but are insignificant for Kenya, indicating differing roles of digital access across countries. Domestic credit and gross capital formation also display mixed signs and significance: some countries (e.g. Kenya) show positive, significant credit effects while others (e.g. Tanzania) show negative or insignificant coefficients. The interest-rate spread tends to be positive and significant in several country regressions, and the reported R-squared values are very high (0.93–0.98), which likely reflects strong trending behavior in the short country series and the small number of observations (23) per country. In short, the country estimates describe country-specific, time-varying associations, but they are not consistent in sign or magnitude across the four East African countries.

In contrast, the panel FMOLS (pooled long-run) estimates provide a clear and consistent long-term perspective across the four countries: LNMCS, LNGDS, and LNDCPS are positive and significant, while LNGCF and LNIRS are strongly negative and significant in the pooled long-term model. The differences stem from three main factors. First, the estimation focus: FMOLS is designed to estimate cointegrating (long-term) relations, while simple OLS for each country captures short-term dynamics, trends, or unique shocks. If the series are non-stationary, OLS regressions can be biased or misleading unless cointegration is considered. Second, pooling versus heterogeneity: FMOLS combines data to estimate an average long-term elasticity, smoothing out individual country differences; therefore, individual country coefficients can legitimately differ (even in sign) from the pooled average. Third, econometric adjustments: FMOLS accounts for endogeneity, serial correlation, and long-term variance, unlike basic OLS time-series models. As a result, panel FMOLS estimates are generally more reliable for long-term analysis. Essentially, country-specific OLS results highlight heterogeneity and short-term trends. However, FMOLS and DOLS panel coefficients should be preferred for understanding the shared long-term relationships across these countries. This is the reason why the robustness checks and tests for cross-sectional dependence are necessary.

Table 12
2SLS and GMM models estimation results.

Dependent variable: Ln HDI						
Panel Two-Stage Least Squares (2SLS)				Panel Generalized Method of Moments (GMM)		
Variable	Coefficient	Prob.	Prob (J-statistic)	Coefficient	Prob.	Prob (J-statistic)
LnMCS	0.18	0.00	0.68	0.18	0.00	0.68
LnGDS	0.25	0.007	0.88	0.25	0.007	0.88
LnIRS	0.18	0.003	0.529	0.18	0.003	0.529
LnGCF	0.55	0.002	0.73	0.55	0.002	0.73
LnDCPS	0.43	0.001	0.88	0.43	0.001	0.88

4.5. Diagnostic tests

The results in Table 14 indicate that the Variance Inflation Factors for all independent variables are well below the threshold value of 10, suggesting that multicollinearity among the variables is not significant enough to necessitate model re-specification.

The Cross-section dependence (CSD) test is performed to check for the presence of cross-sectional dependence in the residuals of panel data models, which is a critical assumption for the reliability of econometric models like FMOLS and DOLS. All four tests provide probability values that are greater than 0.05 as presented in Table 15. This indicates no significant cross-section dependence in both FMOLS and DOLS models. This finding strengthens the reliability of the interpretation of the results, as evidenced by various authors(Gizaw et al., 2024a).

4.6. Comparative context: benchmarking east Africa against other developing regions

To place the East African experience in context, we benchmark our findings against other developing regions. Evidence from South Asia suggests that financial depth, particularly bank credit, significantly enhances human development outcomes through improved income and education (Demirgüç-Kunt et al., 2018). In contrast, Latin America shows more mixed results, where high interest rate spreads and informality constrain the developmental benefits of inclusion (Bruhn, M. & Love, 2014). Compared to these regions, East Africa's unique trajectory is shaped by the rapid adoption of mobile money, which appears to offset structural banking constraints and supports progress in the HDI, particularly in the health and income dimensions(Suri & Jack, 2016). This comparative perspective underlines that while financial inclusion contributes to human development globally, the specific channels differ by region, with digital payments playing a more prominent role in East Africa.

5. Conclusion and policy implications

This study extends financial inclusion indicators to empirically assess their impact on shaping development outcomes. The study's objective is to assess the relationship between economic development proxied by the Human Development Index (HDI) and specific financial inclusion indicators, including mobile cellular subscriptions (MCS), interest rate spread (IRS), gross domestic savings (GDS), and domestic credit to the private sector (DCPS) in Rwanda, Uganda, Tanzania, and Kenya. Additionally, Gross Capital Formation (GCF) is used as a control variable. We employ a panel data set covering 25 years, from 1999 to 2023. The paper uses various econometric techniques, including panel unit root tests, cointegration tests, and fully modified and dynamic ordinary least

Table 13
Country-level regression results.

Dependent variable: LnHDI **Method:** OLS **Observations:** 23 after adjustments								
	Rwanda		**Kenya**		**Uganda**		**Tanzania**	
Variable	**Coef.**	**Prob.**	**Coef.**	**Prob.**	**Coef.**	**Prob.**	**Coef.**	**Prob.**
LnMCS	0.04	0.04	−0.006	0.668	0.04	0.016	0.033	0.0006
LnDCPS	0.002	0.95	0.12	0.022	0.02	0.68	−0.092	0.0079
LnGDS	Negative numbers		0.07	0.006	0.07	0.004	−0.054	0.0164
LnGCF	−0.144	0.16	−0.108	0.111	−0.16	0.032	0.127	0.0001
LnIRS	0.06	0.002	0.037	0.045	0.02	0.233	0.031	0.0000
R-squared	**0.98**		**0.93**		**0.97**		**0.98**	

Table 14
Variance inflation factors.

Variable	Variance	VIF
LNMCS	9.23E-06	7.582071
LNIRS	7.94E-05	7.251131
LNGDS	9.34E-05	5.327179
LNDCPS	0.000126	5.140588
LNGCF	0.000464	8.580072

Table 15
Cross-sectional dependence tests.

	CSD Test in Residuals from FMOLS		CSD Test in Residuals from DOLS	
Test	Statistic	Prob.	Statistic	Prob.
Breusch-Pagan LM	9.796605	0.1335	2.070591	0.9131
Pesaran scaled LM	1.095986	0.2731	−1.134323	0.2567
Bias-corrected scaled LM	1.009029	0.3130	−1.229561	0.2189
Pesaran CD	−0.906420	0.3647	0.198315	0.8428

Null Hypothesis: No cross-section dependence (correlation)
Sample: 1999–2023; Periods included: 24; Cross-sections included: 4.

squares models.

The results of the tests revealed that the variables are non-stationary at "level" and stationary at first differences and integrated with order one. With all variables confirmed to be integrated with the same order, the Pedroni, Kao, Johansen, and Fisher co-integration tests were employed to assess the presence of a long-run relationship or equilibrium among the variables in the model. The findings confirmed that all the variables are cointegrated and have a long-term relationship. This validated the model for exploring long-term relationships and supports the use of dynamic estimation techniques such as Fully Modified OLS (FMOLS) or Dynamic OLS (DOLS) for robust analysis.

The results reported the positive and significant impacts of MCS, GDS, and DCPS on HDI. The magnitude of their effects varies slightly between the models. Conversely, both models agree on the negative impacts of IRS and GCF on HDI. The negative correlation between Gross Capital Formation (GCF) and the Human Development Index (HDI) in East African countries could stem from various economic, structural, and contextual factors. These include investment misallocation, poorly managed projects, negligence of social sectors, with high GCF prioritizing physical infrastructure over critical social spending in health and education, debt-financed investment, lagged effects, and structural economic challenges.

For robustness checks, additional tests were performed to verify the consistency of the results. Mediation path models involving income, health, and education were estimated, with the results shown. Granger causality tests evaluated potential reverse causality between dependent and independent variables. Although FMOLS and DOLS account for endogeneity, we used Two-Stage Least Squares (2SLS) and Generalized Method of Moments (GMM) estimators to go beyond correlation and estimate the causal effect of financial inclusion on HDI while controlling for endogeneity and unobserved heterogeneity. Finally, country-level regressions were conducted to confirm the robustness of the findings.

To ensure the validity and reliability of the results, we examined multicollinearity and cross-sectional dependence. The results reported that the Variance Inflation Factors for all independent variables are well below the threshold value of 10, suggesting that multicollinearity among the variables is insignificant. In addition, the Cross-section dependence (CSD) tests were performed to check for cross-sectional dependence in the residuals of panel data models. The findings indicated no significant cross-section dependence in both FMOLS and DOLS models, thereby strengthening the reliability of the results' interpretation.

This study suggests expanding mobile cellular infrastructure in underserved areas to boost subscription rates and implementing policies to lower mobile service costs for low-income groups. Utilizing mobile technology for financial inclusion through mobile banking and digital payments is vital. Encouraging savings with higher deposit rates, tax incentives, and customized savings plans for rural and low-income populations is essential. Effectively directing savings into productive investments with high developmental returns is critical. Reducing interest rate spreads for SMEs can improve credit access. Strengthening financial literacy programs promotes responsible borrowing and credit use. Promoting innovative financing options like microfinance and cooperative banking can help low-income groups achieve greater financial inclusion. Enhancing capital investments should focus on projects with clear social and economic benefits, such as infrastructure, education, and healthcare, while reinforcing regulatory frameworks for transparency and efficiency. Monetary policies should foster competition among financial institutions to lower lending rates and offer subsidies or guarantees to reduce borrowing costs in strategic sectors.

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Daniel Mburamatare: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Aristide Maniriho:** Writing – review & editing. **Joseph Akumuntu:** Validation, Data curation. **Andre Rukeratabaro:** Software.

Declaration of competing interest

The authors declare no financial conflicts or personal relationships that could influence their work reported in this paper.

Data availability

Data will be made available on request.

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